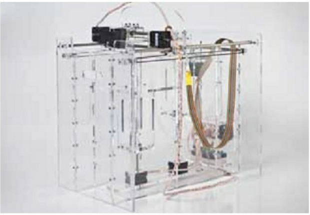
movements of the arm. Apparently, Van As and Owen do not seek any profits from their design—they just want to get enough funds to create prosthetic fingers for whoever needs them, which is why they decided to publish the information in the public domain. The original designs are available at http://chaincrafts.blogspot.in/2012/08/finger-prosthesis-design-details.html, and Liam's hand design is available at http://www.thingiverse.com/thing:44150. The designs will soon be made available in more detailed formats. By publishing these designs in a public forum, they now fall under the concept of 'prior art' (http://en.wikipedia.org/wiki/Prior_art). This means that the developers have given away their right to patent the design, while also ensuring that others cannot patent it, thus keeping it free and public.

New powder-based 3D printing technology

Alex Budding, a researcher at the University of Twente, Singapore, has developed a new rapid-prototyping technology called Pwdr, a powder-based technology that can be extended to enable multi-colour 3D printing. Pwdr is made entirely with off-the-shelf components. The chassis, tool head and electronics can be purchased for less than €1000 and assembled within a few hours using Budding's designs. Instead of using layers of melted plastic, Pwdr uses an HP inkjet cartridge. It deposits a liquid binder, mixed with ink, onto a layer of white gypsum power. Then a roller drags a thin layer of powder across the surface. The process is repeated till the whole object is formed, layer over layer. Then, the object has to be removed from the printer, dusted off and dipped in clear glue that helps solidify it.



The open twist: Pwdr has a maximum build size of 125mm x 125mm x 125mm, a minimum vertical step size of 50 microns, and speed of around 1 minute per layer. It has a pretty decent print resolution of 96 DPI. While there are quite a few open source printer designs already available on the Web, none are powder-based. Budding felt that powder-based printers have a specific application in the field of ceramic materials. His design and code have been open-sourced, and are available on his Github page (http://pwdr.github.com/). While the current design only prints in one colour, the technology could be used to develop a full-colour powder-based printer, thanks to the open design. You could also try to replace the inkjet with a laser, turning the system into a selective laser sintering (SLS) tool. There is also scope to improve the inkjet head, find some way to dispose of the excess powder, or enhance the user-interface. In one of his articles, however, writer Joseph Flaherty issues a word of warning to entrepreneurial readers: "Budding claims that the core patents on this technology expired in 2010, but 3D Systems, ZCorp's parent company, has recently flexed its legal muscles to defend other disputed patents, so perhaps you should consult a patent attorney before putting a model on Kickstarter."

Design a tank with open source META toolkit

The USA's Defense Advanced Research Projects Agency (DARPA) has developed open source software called META that enables anyone to become a tank engineer. The tool will be released shortly, and apparently the framework can achieve the design, manufacture, integration and verification of complex ground vehicles 5X faster than the conventional design-build-test approach. DARPA has also launched a series of contests, the FANG Design Challenges, which encourage engineers to use the toolset to develop vehicles such as an amphibious tank. The details are available on http://www.vehicleforge.org/. The VehicleFORGE website will allow social collaboration around open source vehicle design to compete in the FANG Design Challenges.

The open twist: META is completely open source. It is not based on any one particular approach, metric, technique or tool. The goal of META is to develop model-based design methods for cyber-physical systems that are extremely complex and heterogeneous. It features a structured design flow employing hierarchical abstraction and model-based composition of electromechanical and software components. It further optimises system design, and uses probabilistic formal methods for the system verification, thereby dramatically reducing the need for expensive real-world testing and design iteration. DARPA also aims to develop a component and manufacturing model library for a given airborne or ground vehicle systems domain through extensive characterisation of its features, interactions and properties, right down to the numbered part level. The library will also include context models to reflect various operational environments, and develop a verification flow that generates probabilistic 'certificates of correctness' for the particular cyber-physical system based on stochastic formal methods. END

By: Janani Gopalakrishnan Vikram

The author is a technically-qualified freelance writer, editor and hands-on mom based in Chennai.